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Towards Cardiac Output Estimation Using Earbud Photoplethysmography Sensor

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***Abstract*— Cardiac Output (CO) monitoring is vital for tracking cardiovascular health. In medical settings, cardiac output is monitored invasively, posing physical limitations and making it difficult to monitor CO continuously. In contrast, we leveraged an earbud PPG to non-invasively and continuously monitor CO. Among 11 selected subjects for model development, we observed an MAE of 1.53 L/min and an MAPE of 18.82%. By considering signal quality and PPG morphology, we can improve CO prediction. Our results present an opportunity to lower the barrier for noninvasive PPG-based CO monitoring.**

I. INTRODUCTION

Cardiac output (CO) estimates the volume of blood pumped from the heart to the rest of the body, providing blood flow to vital organs [1]. It plays a crucial role in assessing cardiovascular health and monitoring conditions such as heart failure, shock, and atrial fibrillation [2]. CO is also important for tracking acute physiological changes such as stress responses [2], [3]. CO can be deconstructed into the product of two additional cardiac measures: Stroke Volume (SV) and Heart Rate (HR) [1]. SV describes the volume of blood pumped out of the heart per contraction, while HR describes the number of contractions over a minute.

In medical settings, CO is monitored invasively through pulmonary artery catheterization [4]. As a result of the physical risks and the limits to inpatient setting required for invasive monitoring, researchers have investigated using non-invasive methods to measure cardiac output (CO), notably the feasibility of photoplethysmography (PPG) [5]–[7]. PPG has been well established to be effective for monitoring other cardiac functions such as heart rate, heart rate variability and blood pressure [8], [9]. PPG has also become increasingly integrated into wearable systems for non-invasive and ubiquitous monitoring [10], further supporting its viability as a non-invasive approach for monitoring CO.

Despite its promise, using PPG to monitor CO presents two major challenges. First, PPG is susceptible to noise and motion artifacts. Poor PPG signal quality degrades the accuracy of physiological monitoring due to inaccurate readings [11], [12]. Similarly, the shape of PPG waves (PPG morphology) also affects how well CO can be measured. PPG waves reflect the change of blood volume at a given anatomical acquisition point, similar to how SV reflects the change of blood volume in the heart [13]. Thus, the fidelity with which morphology can be measured affects how well SV and also CO are measured. The detectability of PPG morphology varies based on several factors, notably the sampling rate and anatomical acquisition point. Typically, a higher sampling rate [14] and an acquisition point close to the heart (*e.g.* ear canal [15]) is more ideal for obtaining high-quality PPG waveform morphology measurements. By improving signal acquisition quality, one can more effectively measure PPG morphology related to changes in CO.

In this study, we evaluated the feasibility of monitoring CO using an earbud PPG sensor. We use a research-grade earbud prototype to sample PPG at 450 Hz, much higher than commercial wearables commonly sampled at a much lower frequency (e.g., 25 Hz [16], [17]). We assessed the effect of PPG signal quality and evaluated the influence of waveform morphology on CO prediction. We looked at two waveform morphology conditions: (1) low sampling quality from PPG filtered through a narrow band pass filter (0.7 - 2.1 Hz), simulating morphology typical of lower sampling rates in commercial wearables [18], and (2) high sampling quality from PPG filtered through a wider bandpass filter (0.7 - 3.5 Hz), preserving rich PPG morphology. During the study, participants wore research prototype earbuds while impedance cardiography (ICG) and electrocardiograph (ECG) were collected. ICG was obtained using four band mylar strips completely encircling the neck and torso. ICG and ECG were sampled at 1000Hz and integrated using Biopac hardware and NICO (Noninvasive Cardiac Output Monitoring) and ECG modules [19], [20]. Across a total of 19 subjects among which 11 subjects were useful for our model development, we observed a mean absolute error of 1.53 L/min and of mean absolute percentage error of 18.82% in predicting CO in a leave-one subject-out validation setting, improving CO prediction and performing competitively with existing CO prediction research.

The contributions of our work are described as follows:

- A pipeline to isolate high-quality morphological signals for better CO prediction
- Our approach to filter-out poor quality PPG and identify high-quality PPG waveform estimates more accurate SV and HR using our research earbuds
- Finally, improved SV and HR prediction improves CO prediction, demonstrating the feasibility of estimating CO using a wearable (earbud) PPG system

We contextualize our results with existing non-invasive

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CO prediction research and suggest future directions for exploration.

II. Background and Related Works

PPG has been shown to be a highly versatile physiological signal for capturing various cardiac measurements, including heart rate, heart rate variability, and blood oxygen saturation [21]. In recent years, the integration of PPG sensors into commercial wearable devices has contributed to a significant increase in research on PPG-based cardiac monitoring [10]. As PPG sensors have become more ubiquitous, researchers have dedicated significant effort to developing new sensing algorithms for reliable monitoring, particularly in noisy and variable conditions. Common approaches include assessing signal quality or analyzing the morphological qualities under variable [11], [12], [22]. Continued efforts in this area hold promise for advancing cardiac monitoring applications.

Despite growing interest in using PPG for cardiac monitoring, few studies use PPG to monitor CO. While CO is typically monitored invasively, non-invasive methods are currently gaining traction [23]. Existing PPG-based CO monitoring research uses HRV features [6], PPG morphology [5], [7] and the use of PPG and ECG jointly [24] to develop indices correlated with CO. However, the conditions under which these indices have been developed are confined to lab-based systems and settings where signals are minimally noisy. Although some works show promise, their applicability to noisier wearable systems remains uncertain.

Our work extends the application of PPG-based CO measurement to wearables and addresses the challenges with noisy conditions. We applied signal quality assessments and PPG morphological analyses to improve measurement performance. By taking these steps, we demonstrate the viability of wearable systems for non-invasive CO monitoring.

III. Study Design

We recruited 19 healthy individuals (age range: 19–23; 68.42% female) for our IRB approved study. Participants underwent a five-minute baseline measurement period during which they sat quietly and motionless. Next, they participated in a three-minute speech preparation task and finally a three-minute recovery period. This protocol was intended to induce physiological changes, such as CO variation.

We collected wearable PPG using in-house designed earbuds sampled at 450 Hz at the concha location of the ear (Green LED: 16-213/GHC-YR1S1/3T, Photodetectors: Nisshinbo NJL6193; analog front end: TI AFE44I30). A microcontroller unit (SBS3000 WLCS) triggers the PPG sensing and packages the data stream to a paired smartphone for storage through Bluetooth. We collected the reference physiological data using a BIOPAC system sampled at 1 KHz. The data collected from the earbuds and BIOPAC are synchronized through TTL stimuli sent by BIOPAC. We obtained CO using impedance cardiography using a band-electrode system with two sets of mylar tape applied to the neck and two to the torso. For each non-overlapping minute window, we calculated CO as:

$$CO = SV * HR \tag{1}$$

We used Mindware software to edit and score the ECG and ICG data [25]. SV was estimated using Kubicek's formula:

$$SV = \frac{\rho L^2}{Z_0^2}\frac{dZ(t)}{dt_{max}}T$$

where ρ is the resistivity of the blood, L is the length between inner band electrodes, T is the systolic ejection time, Z_0 is the basal impedance, and $\frac{dZ(t)}{dt_{max}}$ is the maximum rate of impedance change. We scored ICG waveforms using Mindware IMP software. Research assistants scored all the data and a psychophysiologist with decades of experience in analyzing HR, SV, and CO data further verified the scored data [26]. We performed scoring diagnostics to identify physiologically implausible HR, SV, and CO values measured through the BIOPAC system. After cleaning and verifying the data, we used the HR, SV, and CO values derived from the Biopac ICG and ECG as gold-standard reference measurements to evaluate study performance.

To learn a generalizable distribution of CO values, we selected a subset of subjects with a wider individual CO range for analysis. Based on both visual inspection and statistical analysis, we found that some participants had minimal changes in CO in the protocol. Thus, we selected participants with either a $CO_{range} \geq 1.5$ L/min or $CO_\sigma \geq 0.5$ L/min. Of the 19 participants recruited, 11 were selected for analysis. The range of CO values in the subset was 9.432 L/min and the standard deviation was 2.128 L/min. Based on statistics from a review of similar studies ($CO_\sigma = [0.85, 2.1]$) [5], [23], [24] , we found that our study has a sufficient spread of CO values.

IV. Methodology

Through PPG quality assessment and morphological analysis, we hoped to improve CO prediction. Here, we outline our data processing pipeline and prediction process.

A. PPG Preprocessing

We segmented the PPG signals of the earbuds into non-overlapping windows of one minute, aligned to the same window for which HR, SV, and CO values were obtained using BIOPAC. Following segmentation, we filtered the signals to isolate clean PPG. We evaluated two conditions as defined by filtering. We obtained low sampling frequency PPG by applying a narrow bandpass filter (0.7 - 2.1 Hz), removing higher frequency components of the PPG signal. This simulated the morphology typical of lower sampling rates, such as from PPG collected via commercial wearables We obtained high sampling frequency PPG by applying a wider bandpass filter (0.7 to 3.5 Hz) that preserved the richness of the PPG morphology. This was done to simulate both a low quality quality and high quality signal respectively, to interrogate the effect of sampling rate on the morphology quality relevant for estimating CO. In both conditions, non-heartbeat components were removed from the PPG signal (**Figure 1**).

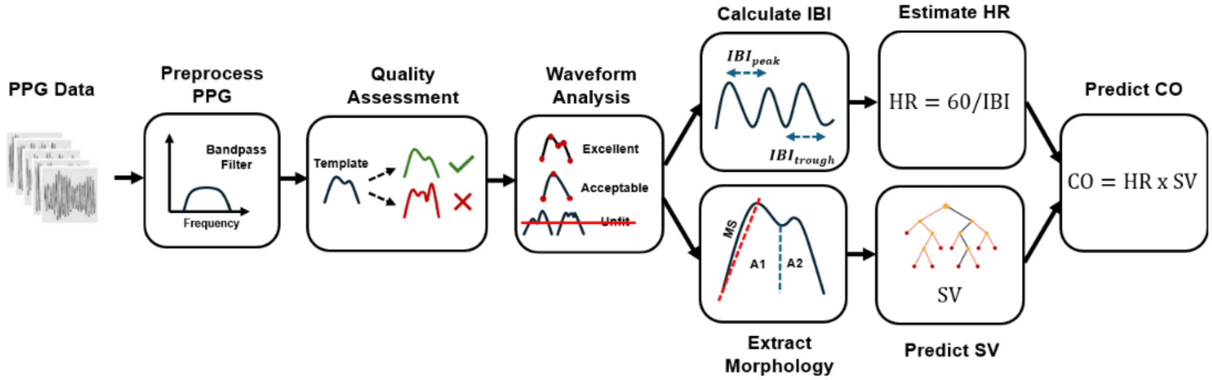

Fig. 1. Algorithms for data preprocessing, PPG signal quality assessment, waveform morphological feature extraction, and cardiac output prediction.

B. PPG Quality Assessment Algorithm

We performed PPG quality assessment to remove noisy waveforms from the analysis. For each PPG segment, we segmented the PPG signal into individual waves by detecting the troughs marking the start and end of each wave. We then employed a template-matching approach to remove low-quality waves following previous approaches derived from the literature [22], [27], [28]. . We resampled all the waveforms to 256 samples in order to standardize comparison of PPG segments, calculated the average of the waves within their given segments, and excluded any waves with a correlation of less than 0.9 with the template wave from the analysis.

C. Waveform Morphology Analysis and Feature Extraction

PPG morphology affects how well cardiac physiology, including CO, can be measured. Our waveform morphology analysis was composed of three major parts: fiducial point detection, waveform classification, and feature extraction.

1) Fiducial Point Detection: A key component of morphological analysis is the detection of fiducial points — landmarks marking relevant physiological events. Fiducial points in the original wave (**Figure 2**) include the *onset*, systolic peak (*sys*), dicrotic notch (*dic*), diastolic peak (*dia*), and *end*. Fiducial points in the second wave only include the maximum upslope of the wave (*ms*). Fiducial points in the second derivative include the early systolic positive wave (*a*), the early negative systolic wave (*b*), late re-increasing systolic wave (*c*), late re-decreasing systolic wave (*d*), and the dicrotic notch wave (*e*). Fiducial points in the third derivative include the early systolic component (p_1) and the late systolic component (p_2) These fiducial points can be seen in **Figure 2**. We detected fiducial points for each wave by calculating the zero-crossing points of a given wave's derivative [29].

2) Waveform Classification: We classified individual waves into three categories: "Excellent", "Acceptable", and "Unfit". This classification system was originally proposed by [12]. "Excellent" waves contained both systolic and diastolic morphologies, enabling the detection of all original wave fiducial points (*onset, sys, dic, dia, end*). "Acceptable" waves included only systolic morphology (*onset, sys, end*), making the dicrotic notch (dic) and diastolic peak (dia) undetectable **(Figure 2)**. "Unfit" waves lacked discernible morphology, preventing reliable fiducial point detection (see Figure 1 - *Waveform Analysis*). They possessed unrealistic morphology such as pronounced diastolic morphology ($y(dia) > y(sys)$) or more than two visible peaks making it difficult to discern the systolic and diastolic morphology. Unfit waves were excluded from the analysis.

TABLE I
MORPHOLOGICAL FEATURES WHICH CAN BE EXTRACTED FROM PPG BASED ON WAVE QUALITY

Source	Quality	Feature	Definition
Amplitude	Excellent	AI	$\frac{y(p_2)-y(p_1)}{y(sys)-y(onset)}$
		RI	$\frac{y(dia)-y(0)}{y(sys)-y(0)}$
		RI_{p1}	$\frac{y(dia)-y(0)}{y(sys)-y(p1)}$
		RI_{p2}	$\frac{y(dia)-y(0)}{y(sys)-y(p2)}$
		$ratio_{p1,p2}$	$\frac{y(p2)-y(0)}{y(p1)-y(0)}$
		$d-a$	$y''(d)-y''(a)$
		$e-a$	$y''(d)-y''(a)$
		AGI	$\frac{y''(b)-y''(c)-y''(d)-y''(e)}{y''(a)}$
	Acceptable & Excellent	Pulse Amp.	$y(sys)-y(onset)$
		MS Amp.	$y'(ms)-y(onset)$
		$b-a$	$y''(b)-y''(a)$
		$c-a$	$y''(c)-y''(a)$
Area & Slope	Excellent	$A1$	$A(t(onset),t(dic))$
		$A2$	$A(t(dic),t(end))$
		IPA	$A2/A1$
		$IPAD$	$A2/A1+d-a$
	Acceptable & Excellent	MS	$\frac{y'(ms)}{y(sys)-y(onset)}$

3) Morphological Feature Extraction: We extracted morphological features for each wave, including amplitude, area, and slope features, given their relationship to SV [13]. We extracted features per waveform based on how they were classified. A list of these features for both acceptable and excellent quality PPG signals are available in Table I. A represents area, y represents amplitude, the $'$ symbol represents derivative, and t represents timing. Additional details about

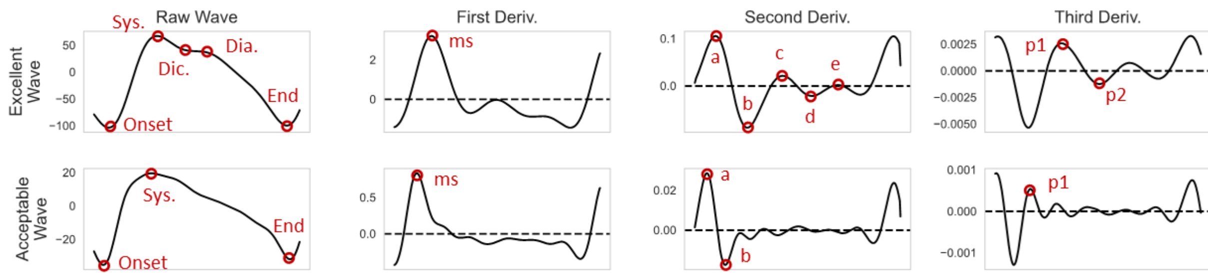

Fig. 2. Acceptable vs. Excellent waveforms and corresponding fiducial points from the original PPG signal, their first, second, and third derviative.

these features can be found in [18], [30]. Both acceptable and excellent features features are extractable from excellent PPG waves, but only acceptable features are extractable from acceptable PPG waves. We computed features per each minute-long PPG segment by calculating the mean feature value cross all waves for the given segment.

D. Cardiac Output Prediction

For each non-overlapping minute window, we calculated CO using equation (1). This required estimating HR and predicting SV first. HR and SV are calculated as follows.

We evaluated the results by computing the mean absolute error (MAE) and mean absolute percent error (MAPE) in reference to the HR values obtained using the BIOPAC.

1) SV Prediction: We predicted SV using random forest models trained on the extracted morphological features. We trained one random forest model on low sampling quality (0.7 - 2.1 Hz bandpass) PPG data. We trained a second random forest model on high sampling quality (0.7 - 3.5 Hz bandpass) PPG data. We trained the two models to determine if rich waveform morphology from higher sampling quality would produce better performance than morphology from lower sampling quality similar to that of commercial wearables. Both models used the same parameters (estimators = 100, max depth = 4), and were validated in a leave-one-subject-out validation setting. We assessed the performance using MAE and MAPE with respect to the reference SV values. We also calculated Pearson's r to qualitatively assess the fit of the model.

2) CO Prediction: We multiplied the estimated HR and predicted SV values to obtain CO value predictions for both high sample quality and low sample quality conditions. Once again, we assessed the performance using MAE and MAPE in relation to the reference CO values, and Pearson's r to qualitatively evaluate the model's fit. We compare our results to other existing works via MAPE, as recommended by Critchley *et al.* [23].

V. Results

A. Heart Rate Estimation

Per HR estimation, we found that removing unfit and low-quality PPG waves (MAE = 0.78, MAPE = 1.31%) improved performance over not excluding those waves (MAE = 1.19, MAPE = 1.99%). We also found our approach outperformed Dynamic Fine Filtering (DFF) (MAE = 1.47, MAPE = 2.47%) regardless of whether or not we excluded low-quality waves. Based on the improved results, we took the HR estimated from our approach of using IBI_{trough} on quality-assessed PPG waves to predict CO.

TABLE II
HR Estimation, SV Prediction, and CO Prediction Results.
* See Sec. V-C for comments about this result

	MAE ↓	MAPE (%) ↓
HR Estimation (Sec. V-A)		
Dynamic Fine Filter [28]	1.47	2.47%
Trough, No Quality Assessment	1.19	1.99%
Trough, Quality Assessment	0.78	1.31%
SV Prediction (Sec. V-B)		
Low Sampling Quality	32.94	24.59%
High Sampling Quality	23.42	17.49%
CO Prediction (Sec. V-C)		
Lee *et al.* [6]	-	47%
Wang *et al.* - PAT [24]	-	23.43%
Wang *et al.* - IHAR [5]	-	16.2%*
Low Sampling Quality	2.23	25.72%
High Sampling Quality	1.53	18.82%

B. Stroke Volume Prediction

Per SV prediction, we found the model trained on high sampling quality morphology (MAE = 23.42, MAPE = 17.49%, r = 0.62) improved performance over the model trained on low sampling quality morphology (MAE = 32.94, MAPE = 24.59%, r = 0.12). We calculate CO with SV predicted from both high sampling quality and low sampling quality to assess the magnitude of improvement high sampling quality morphology provides.

C. Cardiac Output Prediction

We found CO calculated from high sampling quality morphology (MAE = 1.53, MAPE = 18.82%, r = 0.66) outperformed CO calculated from low sampling quality morphology (MAE, 2.23, MAPE = 25.72% , r = 0.24). Our study improved CO prediction compared to Lee *et al.* [6] by using an enhanced set of morphology features compared to their use of mainly HRV features, suggesting morphology is more informative for CO prediction. Our results were comparable to Wang *et al.*'s [5], though we were unable to implement IHAR. IHAR, defined as the ratio of NHA (normalized harmonic area) to IPA (inflection point area), is a morphological feature that requires the presence of multiple harmonics in the frequency domain. However, we found PPG waves only exhibited a single harmonic, preventing the implementation of the IHAR feature as described.

VI. Discussion

Our results present the trade-off between high and low sampling frequency signals. In particular, when high sampling frequency signals are used, they retain additional morphological features in the PPG compared to low-frequency signals. We see the downstream effect of using a high sampling frequency signal in terms of improved CO prediction. Previous work [14] conducted experiments to systematically downsample the PPG signal to identify the minimal sampling frequency for the best quality biomarkers with the minimal sampling rate for their application (e.g., heart rate variability monitoring). Future work can replicate such experiments to establish the minimal sampling frequency to achieve the best quality cardiac output estimation.

Though higher sampling frequency data might not be feasible to collect continuously in a commercial device due to the impact of the sensor on the battery life of the device, select usage of short-term windows (*e.g.* 60 seconds in our work) can be used to collect some amount of data for CO estimation. We provide these insights for researchers or designers developing passive CO solutions with PPG.

Furthermore, due to the limited size of our dataset, we did not investigate any deep learning modeling techniques. Given the smaller dataset we used for evaluation, we employed classical signal processing and machine learning methods and incorporated domain knowledge to assess feasibility. The encouraging results suggest further exploration in larger datasets with noisier data, where more complex methods, such as a teacher-student model that embeds similar domain knowledge, could effectively isolate CO-related signals.

VII. Conclusion

We demonstrated the feasibility of noninvasive cardiac output (CO) monitoring using a prototype earbud. By using quality filters to remove low-quality PPG segments, and using high sampling quality morphology enabled by our earbuds, we developed a model that showed superior performance over the low-sampling quality signal and comparable performance against other PPG systems. Our study provides insights on the trade-offs between the low-sampling and high-sampling quality PPG for continuous CO monitoring using consumer wearables.

References

[1] J. King and D. R. Lowery, “Physiology, Cardiac Output,” in *StatPearls*, 2017.

[2] F. Vancheri, G. Longo, E. Vancheri, and M. Y. Henein, “Mental Stress and Cardiovascular Health,” *Journal of Clinical Medicine*, 2022.

[3] J. Blascovich and W. B. Mendes, “Challenge and threat appraisals: The role of affective cues,” in *Feeling and thinking: The role of affect in social cognition*, 2000.

[4] L. Mathews and K. R. Singh, “Cardiac output monitoring,” *Annals of Cardiac Anaesthesia*, 2008.

[5] L. Wang, E. Pickwell-MacPherson, Y. Liang, and Y. Zhang, “Noninvasive cardiac output estimation using a novel photoplethysmogram index,” in *EMBC, IEEE*, 2009.

[6] Q. Y. Lee *et al.*, “Estimation of cardiac output and systemic vascular resistance using a multivariate regression model with features selected from the finger photoplethysmogram and routine cardiovascular measurements,” *BioMedical Engineering OnLine*, 2013.

[7] W.-H. Lin, X. Li, Y. Li, G. Li, and F. Chen, “Investigating the physiological mechanisms of the photoplethysmogram features for blood pressure estimation,” *Physiological Measurement*, 2020.

[8] G. Fortino and V. Giampà, “PPG-based methods for non invasive and continuous blood pressure measurement: an overview and development issues in body sensor networks,” in *IEEE International Workshop on Medical Measurements and Applications*, 2010.

[9] N. Pinheiro *et al.*, “Can PPG be used for HRV analysis?” in *EMBC, IEEE*, 2016.

[10] P. H. Charlton *et al.*, “Wearable Photoplethysmography for Cardiovascular Monitoring,” *Proceedings of the IEEE*, 2022.

[11] J. A. Sukor, S. J. Redmond, and N. H. Lovell, “Signal quality measures for pulse oximetry through waveform morphology analysis,” *Physiological Measurement*, 2011.

[12] M. Elgendi, “Optimal Signal Quality Index for Photoplethysmogram Signals,” *Bioengineering*, 2016.

[13] W. B. Murray and P. A. Foster, “The peripheral pulse wave: Information overlooked,” *Journal of Clinical Monitoring*, 1996.

[14] M. D. Peláez-Coca, A. Hernando, J. Lázaro, and E. Gil, “Impact of the PPG Sampling Rate in the Pulse Rate Variability Indices Evaluating Several Fiducial Points in Different Pulse Waveforms,” *IEEE Journal of Biomedical and Health Informatics*, 2022.

[15] K. Budidha and P. A. Kyriacou, “Devepopment of an optical probe to investigate the suitability of measuring photoplethysmographs and blood oxygen saturation from the human auditory canal,” in *EMBC, IEEE*, 2013.

[16] C. Kryder, “How Does the Oura Ring Track My Sleep?” 2022.

[17] A. G. Polak, B. Klich, S. Saganowski, M. A. Prucnal, and P. Kazienko, “Processing Photoplethysmograms Recorded by Smartwatches to Improve the Quality of Derived Pulse Rate Variability,” *Sensors*, 2022.

[18] L. Zhang *et al.*, “Morphological Photoplethysmography Features Enhance Stress Detection in Earbud Sensors,” in *EMBC, IEEE*, 2024.

[19] “ICG Impedance Cardiography Noninvasive Cardiac Output Module.” [Online]. Available: https://www.biopac.com/product/noninvasive-cardiac-output-module/

[20] “ECG: Cardiology | Research | BIOPAC.” [Online]. Available: https://www.biopac.com/application/ecg-cardiology/

[21] M. A. Almarshad, M. S. Islam, S. Al-Ahmadi, and A. S. BaHammam, “Diagnostic Features and Potential Applications of PPG Signal in Healthcare: A Systematic Review,” *Healthcare*, 2022.

[22] C. Orphanidou, “Quality Assessment for the Photoplethysmogram (PPG),” in *Signal Quality Assessment in Physiological Monitoring: State of the Art and Practical Considerations*, C. Orphanidou, Ed. Springer International Publishing, 2018.

[23] L. A. H. Critchley and J. A. J. H. Critchley, “A Meta-Analysis of Studies Using Bias and Precision Statistics to Compare Cardiac Output Measurement Techniques,” *Journal of Clinical Monitoring and Computing*, 1999.

[24] L. Wang, C. C. Y. Poon, and Y. T. Zhang, “The non-invasive and continuous estimation of cardiac output using a photoplethysmogram and electrocardiogram during incremental exercise,” *Physiological Measurement*, 2010.

[25] “Analysis Software - MindWare Technologies,” 2020. [Online]. Available: https://www.mindwaretech.com/analysis-software/

[26] W. B. Mendes, “Assessing autonomic nervous system activity,” in *Methods in social neuroscience*, 2009.

[27] P. K. Lim *et al.*, “Adaptive template matching of photoplethysmogram pulses to detect motion artefact,” *Physiological Measurement*, 2018.

[28] R. K. Sah *et al.*, “Heart Rate Variability Estimation with Dynamic Fine Filtering and Global-Local Context Outlier Removal,” in *ICASSP, IEEE*, 2024.

[29] F. Foroozan, M. Mohan, and J. S. Wu, “Robust Beat-To-Beat Detection Algorithm for Pulse Rate Variability Analysis from Wrist Photoplethysmography Signals,” in *ICASSP, IEEE*, 2018.

[30] E. Mejía-Mejía, J. Allen, K. Budidha, C. El-Hajj, P. A. Kyriacou, and P. H. Charlton, “Photoplethysmography signal processing and synthesis,” in *Photoplethysmography*, 2022.